Multimodal Prompt Tuning With Modified Knowledge Distillation for the Joint Classification of Hyperspectral Image and LiDAR Data

Yi Kong, *Member, IEEE*, Shaocai Yu, Yuhu Cheng, *Senior Member, IEEE*, C. L. Philip Chen, *Life Fellow, IEEE*, and Xuesong Wang, *Member, IEEE*

***Abstract*—While pretrained foundation models (FMs) exhibit remarkable generalization across diverse tasks, their full fine-tuning under limited labeled data remains prohibitively expensive and prone to performance degradation or overfitting. To mitigate these challenges in the context of multimodal remote sensing image classification, we propose multimodal prompt tuning with modified knowledge distillation (MPT-MKD), a parameter-efficient learning framework enabling accurate class predictions while updating only a small fraction of model parameters. Prompt tuning is adapted to multimodal remote sensing by injecting modality-specific learnable prompts (spectral and light detection and ranging (LiDAR) prompts) into a student FM, whereby cross-modal interactions are captured through self-attention. In addition, a knowledge distillation (KD) scheme is designed that leverages both limited labeled samples and abundant unlabeled data, where a teacher FM, with parameters frozen except for a learnable classifier, generates soft pseudo-labels for the student FM. To mitigate potential inaccuracies in the logits predicted by the teacher FM, a similarity- and class-consistency-based selection strategy is employed to identify a more reliable subset of unlabeled samples and to reconstruct modified soft targets, thereby alleviating the adverse effects of noisy distillation signals. Experimental results on three real hyperspectral image (HSI) and LiDAR datasets demonstrate the effectiveness of the proposed method.**

***Index Terms*—Classification, foundation model (FM), hyperspectral image (HSI), light detection and ranging (LiDAR), prompt tuning.**

I. Introduction

MULTIMODAL remote sensing images refer to observations acquired by multiple heterogeneous sensors [1]. For example, hyperspectral images (HSIs) contain numerous spectral bands that can reveal subtle differences between different classes of ground objects [2], while light detection and ranging (LiDAR) data can reflect the elevation information of ground objects [3], [4]. Therefore, how to extract fused and discriminative features from multimodal remote sensing images and achieve higher pixel-wise classification performance is currently one of the research hotspots in the field of remote sensing [5], [6].

The representative multimodal remote sensing image fusion and classification methods are those based on handcrafted features and those based on deep learning (DL) [7], [8]. Among them, handcrafted feature-based methods typically involve two independent steps: the extraction and fusion of multimodal features, followed by a classification procedure. However, these methods not only rely on expert knowledge but also tend to extract features limited to one or a few aspects, making it challenging to fully capture the inherent characteristics of ground objects. In contrast, DL methods can automatically extract high-level, abstract, and compact features directly from raw data with minimal human intervention [9], [10].

The most commonly used DL methods are mainly based on convolutional neural network (CNN) [11] and vision transformer (ViT) [12]. Due to the inherent mechanisms of CNNs, such as local receptive fields and weight sharing, CNN-based methods often focus on extracting local features. For example, Zhang et al. [13] proposed a patch-to-patch CNN that uses a multibranch CNN architecture to extract multiscale features from multisource remote sensing images. Zhao et al. [14] accomplished the joint classification of HSI and LiDAR through a hierarchical random walk network, which integrates a dual-tunnel CNN architecture to capture spectral and spatial features from different data sources. Song et al. [15] employed convolutional operations to capture the frequency-domain features of HSI and LiDAR, thereby extracting deeper spectral–spatial representations. Subsequently, a cross-convolution operation was used to facilitate the interaction and fusion of HSI and LiDAR features. Ma et al. [16] first divided the extracted features into high- and low-level components, and then employed a group convolutional attention mechanism to enable their interaction and fusion.

Compared to CNNs, ViT-based methods often focus on extracting global features. Given multimodal remote sensing data, different kinds of tokenization operations are first performed to obtain a series of visual tokens. Typically, each token contains small local features of a ground object (such as spectral, spatial, or spectral–spatial features). Subsequently,

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the self-attention or cross-attention mechanisms compute the attention scores based on the similarities among different tokens, facilitating the weighted aggregation of global features [17], [18]. Xue et al. [19] employed a spectral sequence transformer to capture the long-range dependencies in the spectral dimension of HSI and utilized a multilayer transformer to extract spatial features from both HSI and LiDAR data. Additionally, multimodal features were fused using a cross-attention mechanism. Roy et al. [20] introduced a learnable spectral and spatial morphological network, utilizing both spectral and spatial morphological convolution operations, coupled with the attention mechanism. This design enhances the interplay between the structural and shape information of the HSI token and the CLS token. Dong et al. [21] introduced self- and cross-guided attention at each layer to make the model focus on regions and channels of interest, while a spectral supplementary module is also designed to obtain representations of central pixels with stronger discriminative ability. Yao et al. [22] utilized parallel branches equipped with the separable convolution modules to capture spatial and modality-specific channel features, while enhancing the interaction between different data sources through a cross-modality attention mechanism. Yang et al. [23] proposed a transformer-based heterogeneous saliency graph representation method. This method first employs a heterogeneous graph encoder to extract non-Euclidean structural features, and then introduces a multiconvolution modulator without self-attention to model long-range dependencies. Finally, a mean-forward strategy is adopted to alleviate the overfitting problem. Fu et al. [24] proposed a spatial grouping constraint method and a spectral weighting constraint module to extract multiscale features in the spatial domain and to enhance semantic features in the spectral dimension, respectively. Besides, Yang et al. [25] pioneered the integration of text supervision and multidimensional contrastive learning for HSI and LiDAR classification, offering a novel approach to leverage multimodal data semantics. Feng et al. [26] integrated CNN-based feature extraction with an enhanced Transformer architecture to achieve deep and contrastive cross-modal fusion of hyperspectral and LiDAR data. Additionally, some methods comprehensively leverage the strengths of the two mainstream DL models by jointly employing CNN and ViT [27], [28]. For example, Wang et al. [29] proposed a multiscale pyramid fusion framework based on spatial–spectral cross-modal attention, which effectively enhances HSI-LiDAR data fusion and classification performance through multiscale feature extraction and cross-modal information interaction. Wang et al. [30] employed a triple-branch CNN to jointly extract spatial, spectral, and elevation features from HSI and LiDAR data and introduced hierarchical and modality attention mechanisms to achieve effective cross-modal feature fusion.

Currently, the vast majority of DL methods need to be trained from scratch for specific downstream tasks, which not only requires carefully designed network architectures but also a large amount of labeled samples for support. Fortunately, in recent years, several foundation models (FMs) for remote sensing imagery have been proposed. These models are typically based on transformers, utilize self-supervised learning algorithms, and are trained on large-scale remote sensing datasets. FMs can extract generalized features and demonstrate promising performance across a variety of downstream tasks, garnering increasing research attention. Sun et al. [31] collected a large-scale dataset of two million remote sensing images from satellite and aerial platforms to support the training of FMs. Additionally, to address the issue of information loss in small objects that often occurs in conventional mask image modeling-based self-supervised learning methods, an incomplete patch mask mechanism is implemented, which involves randomly preserving certain pixels during the masking process. EarthGPT [32], a multimodal large language model, was designed for comprehensive image interpretation across multiple remote sensing modalities. It utilizes a visual-enhanced perception mechanism, a cross-modal mutual comprehension approach, and a unified instruction tuning method to effectively integrate visual and linguistic data for enhanced performance. Liu et al. [33] designed a language-vision model for remote sensing images, aligning visual features with semantic features in a common feature subspace, thereby enabling the extraction of remote sensing data features with strong semantics. However, most of the aforementioned FMs are tailored to handling RGB images and struggle to model the relationships between different spectral bands in multispectral images. Therefore, Hong et al. [34] proposed SpectralGPT, which includes a 3-D masking mechanism for processing spectral data, an encoder for extracting image features, and a decoder for multiobject reconstruction. Additionally, to enhance the model's generalization capability while processing images at different scales, a progressive pretraining strategy is also introduced. Subsequently, Wang et al. [35] designed HyperSIGMA, a ViT-based FM for HSI interpretation. They first collected a large-scale hyperspectral dataset, HyperGlobal. Then, a novel sparse sampling attention mechanism and a spectral enhancement model are designed to further enhance the FM's capabilities, thereby achieving outstanding performance on various HSI interpretation tasks.

However, most existing HSI-LiDAR fusion methods that are based on cross-attention mechanisms or hybrid CNN–transformer architectures are trained from scratch. This training paradigm not only requires multiple rounds of model retraining to obtain relatively optimal hyperparameters but also tends to produce task-specific features with limited generalization ability across different downstream scenarios. In contrast, recent advances in large spectral FMs have demonstrated strong generalization capabilities across diverse tasks by learning universal representations from large-scale HSIs. To this end, how to fully exploit the potential of FMs and achieve superior performance through tuning only a minimal number of parameters has gradually become a trending research topic [36]. The visual version evolved from prompt learning in the natural language processing field, known as visual prompt tuning (VPT) [37], enables model generalization to specific downstream tasks by adding extra learnable tokens to the basic blocks of the ViT. The VPT is a parameter-efficient tuning method that keeps the parameters of the pretrained FM frozen, with learnable parameters limited to the newly added tokens. This not only significantly reduces the fine-tuning time and computational resources but also avoids the risk of network degradation. Therefore, inspired by the VPT, we introduce a multimodal PT method for the joint classification

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of HSI and LiDAR data. The complementary information is first encoded into the HSI prompts and LiDAR prompts. Then, the information interaction between multimodal remote sensing images can be achieved through the self-attention mechanism in the basic blocks of ViT. Additionally, a series of extra added prompts will bring additional computational burden on the FM. Inspired by the mixture of experts (MoE), we design a selection and mixing mechanism that first chooses a subset from a set of prompts, which is more beneficial for downstream classification tasks. Then, the selected prompts will be mixed into a single prompt, thereby reducing both the computational costs and the impact of noisy pixels.

Typically, the newly added parameters are optimized by supervised learning algorithms on a limited set of labeled samples, which may suffer from the risk of overfitting. To this end, inspired by work [38], we introduce the knowledge distillation (KD) framework to transfer knowledge from a larger teacher FM to a smaller student FM by aligning the predicted logits on extensive unlabeled samples. However, the logits predicted by the teacher model may be inaccurate, which could mislead the learning of the student model. To this end, we select a more reliable subset from the logits corresponding to the labeled samples to reconstruct a modified logit. In summary, we propose a joint classification of HSI and LiDAR data method named multimodal prompt tuning with modified KD (MPT-MKD). The main contributions include the following.

1) A parameter-efficient tuning method for FMs has been designed to facilitate the joint classification of HSI and LiDAR data. HSI and LiDAR prompt vectors (PVs) can be obtained through different prompt learners and further enable feature interaction via the self-attention mechanism.
2) To leverage the limited labeled samples and a large number of unlabeled samples, the KD is constructed using FMs of different sizes, with a larger FM serving as the teacher model and a smaller FM as the student model.
3) A novel MKD strategy is designed, which involves selecting a subset from the labeled samples to construct the modified logits, thereby reducing the negative impact of inaccuracies in the pseudo-supervision information on the learning process of the student model.
4) To reduce ineffective feature fusion caused by noisy pixels and to alleviate the increased computational load due to newly added PVs, a prompt mixing method is introduced. A self-gating mechanism is employed to identify more discriminative pixels within the local regions. The corresponding PVs of these pixels are then weighted and mixed to obtain a single PV.

II. MPT-MKD-Based Multimodal Classification

As shown in Fig. 1, the proposed MPT-MKD primarily consists of six steps.

1) Training a teacher FM with limited labeled samples and obtaining the predicted logits of labeled and unlabeled samples. In this procedure, the parameters of the basic blocks are frozen and an additional classifier is learnable.
2) Obtaining the spectral prompts through spectral prompt learner.

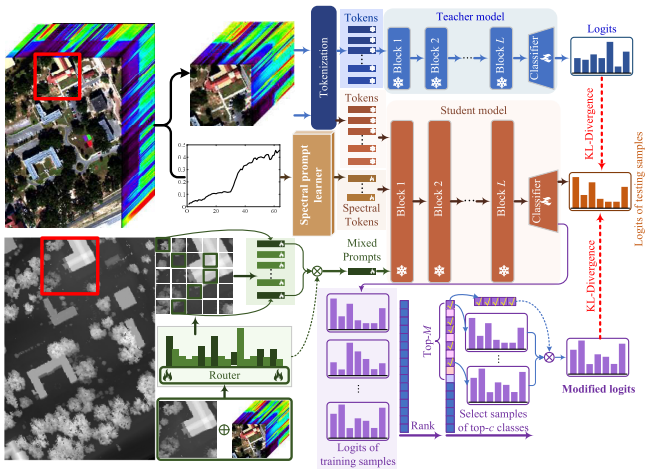


Fig. 1. Flowchart of MPT-MKD.

3) Splitting the local region representations into patches and mapping each of them into the embedding space.
4) Compute the score for each patch using a gating network, and perform a weighted mixture of the embeddings to obtain the mixed prompts.
5) Input the spectral prompts, the mixed prompts, and the tokens into the blocks of the student FM for feature extraction, subsequently obtaining the predicted logits for labeled samples and unlabeled samples.
6) For each unlabeled sample's logit, identify the top-M (M most similar) samples from the training set and the top-c (count the c most frequently occurring) classes among them.

Then, the modified logit for each unlabeled sample is obtained by a weighted sum of the predicted logits of the top-c classes from the top-M labeled samples. The final learning objective can be constructed by combining the conventional KD loss (Kullback–Leibler (KL) divergence between the output logits of teacher and student models), the modified KD loss (KL divergence between the student's output and the modified vectors), and the cross-entropy loss.

A. Teacher Model Training

Given the original HSI $\boldsymbol{X}^{(\mathrm{HSI})} \in \mathbb{R}^{H\times W\times D}$ and LiDAR data $\boldsymbol{X}^{(\mathrm{LiDAR})} \in \mathbb{R}^{H\times W\times d}$, where H and W are the width and height of HSI and LiDAR data, and D and d represent the number of bands of HSI and LiDAR data, respectively, the local neighborhood representations $\tilde{\boldsymbol{X}}^{(\mathrm{HSI})} \in \mathbb{R}^{h\times h\times D}$ and $\tilde{\boldsymbol{X}}^{(\mathrm{LiDAR})} \in \mathbb{R}^{h\times h\times d}$ for HSI and LiDAR can be constructed by selecting several pixels around a certain pixel, where h represents the height and width of the local neighborhood representations.

Remote sensing FMs are trained on large-scale remote sensing images with numerous learnable parameters, enabling them to exhibit general feature extraction capabilities. FMs are generally based on the transformer architecture due to its scalability and flexibility. By combining FMs with parameter-efficient fine-tuning techniques, they can be readily generalized to a range of downstream tasks, such as pixel-wise classification, semantic segmentation, etc. Among the open-access FMs, HyperSIGMA [28] is specifically designed for HSI,

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offering three different versions with various model sizes (base, huge, and large). In this article, we consider a variant of HyperSIGMA (named SpatSIGMA), due to its efficiency and comparable accuracy, which also provides three pretrained models with different capacities. Typically, larger FMs are able to learn more knowledge, thereby possessing stronger feature representation capabilities. Therefore, we can consider the larger FMs as teacher models and the smaller ones as student models. Then, a linear classifier $f_{\boldsymbol{\theta}^{(\mathrm{C})}}(\cdot)$ with learnable parameters $\boldsymbol{\theta}^{(\mathrm{C})}$ is added to the head of the teacher model to obtain the predicted logits of the labeled samples

$$\begin{cases} \boldsymbol{F} = f_{\boldsymbol{\theta}^{(\mathrm{T})}}\left(\tilde{\boldsymbol{X}}^{(\mathrm{HSI})}\right) \\ \tilde{\boldsymbol{y}}^{(\mathrm{T})} = f_{\boldsymbol{\theta}^{(\mathrm{C})}}(\boldsymbol{F}) \end{cases} \tag{1}$$

where $\boldsymbol{\theta}^{(\mathrm{T})}$ denotes the parameters of the teacher model, which is always frozen during the subsequent training procedure. Then, $\boldsymbol{\theta}^{(\mathrm{C})}$ can be optimized by minimizing the standard cross-entropy CE$(\cdot)$ loss over the whole training set as follows:

$$\boldsymbol{\theta}^{(\mathrm{C}),*} = \arg\min_{\boldsymbol{\theta}^{(\mathrm{C})}} -\frac{1}{N^{\mathrm{tr}}} \sum_{i=1}^{N^{\mathrm{tr}}} \boldsymbol{y}_i \log \tilde{\boldsymbol{y}}_i \tag{2}$$

where N^{tr} represents the number of training samples and $\boldsymbol{y}$ denotes the ground truth of the training samples. Then, the logits $\tilde{\boldsymbol{y}}_{\mathrm{TE}}$ corresponding to the unlabeled samples predicted by the teacher model are

$$\tilde{\boldsymbol{y}}_{\mathrm{TE}} = f_{\boldsymbol{\theta}^{(\mathrm{C}),*}}\left(f_{\boldsymbol{\theta}^{(\mathrm{T})}}\left(\tilde{\boldsymbol{X}}_{\mathrm{TE}}^{(\mathrm{HSI})}\right)\right) \tag{3}$$

where $\tilde{\boldsymbol{X}}_{\mathrm{TE}}^{(\mathrm{HSI})}$ denotes the unlabeled samples of HSI.

B. Prompt Learners

The student model is constructed inspired by the paradigm of VPT [37], which is a parameter-efficient fine-tuning method that improves model performance on specific downstream tasks by adding a small number of learnable parameters, known as the PVs, in the input space. For the task of joint classification of HSI and LiDAR, the information of both modalities should be considered in the computation process of the PVs. For HSI, we can obtain the PVs through a prompt learner, which can be either a multilayer perceptron (MLP) or a CNN. For the convenience of obtaining multiple PVs, a 1D-CNN is selected to construct the spectral prompt learner. The spectral PVs are calculated as

$$\boldsymbol{P}^{(\mathrm{spe})} = f_{\boldsymbol{\theta}^{(\mathrm{SPL})}}\left(\boldsymbol{x}^{(\mathrm{HSI})}\right) \tag{4}$$

where $\boldsymbol{x}^{(\mathrm{HSI})}$ represents the spectral features corresponding to each pixel and $\boldsymbol{\theta}^{(\mathrm{SPL})}$ is the learnable parameters of spectral prompt learner. For the modality of LiDAR data, the corresponding PVs should focus on extracting spatial features. Therefore, a 2D-CNN is selected as the LiDAR prompt learner, and the corresponding PVs $\boldsymbol{P}^{(\mathrm{LiDAR})}$ can be calculated as

$$\boldsymbol{P}^{(\mathrm{LiDAR})} = f_{\boldsymbol{\theta}^{(\mathrm{2D})}}\left(\tilde{\boldsymbol{X}}^{(\mathrm{LiDAR})}\right) \tag{5}$$

where $\boldsymbol{\theta}^{(\mathrm{2D})}$ denotes the parameters of the LiDAR prompt learner and $\boldsymbol{P}^{(\mathrm{LiDAR})} \in \mathbb{R}^{h\times h\times d^{(\mathrm{LiDAR})}}$. Specifically, here we choose equal-width 2-D convolutions to achieve dimension consistency in the spatial domain. Consequently, each pixel after convolution is equivalent to a weighted aggregation of pixels in a local region. Further considering the implications, if all h^2 LiDAR PVs are added into the input space of the student FM, two issues may arise. On the one hand, the quadratic complexity of self-attention computations will lead to a significant increase in computational cost. On the other hand, while some pixels carry discriminative elevation information, others may provide invalid or even detrimental information. In response to these challenges, an MoE-inspired method is proposed, which uses a weighted mixture of multiple LiDAR PVs to form a single PV.

The above h^2 LiDAR PVs are considered h^2 experts, each carrying the elevation features corresponding to one pixel. Then, a gating mechanism is utilized to obtain the activation $\boldsymbol{g}_i$ corresponding to each pixel

$$\boldsymbol{g}_i = G_{\boldsymbol{\theta}^{(\mathrm{Gate})}}\left(\left[\tilde{\boldsymbol{X}}_i^{(\mathrm{HSI})}, \tilde{\boldsymbol{X}}_i^{(\mathrm{LiDAR})}\right]\right) \tag{6}$$

where $\boldsymbol{g}_i \in \mathbb{R}^{h^2}$ denotes the output of the gating network. Subsequently, the probability of selecting each expert $\boldsymbol{p}_i$ is determined as

$$\begin{cases} \boldsymbol{p}_i = \mathrm{soft\,max}\left(\tilde{\boldsymbol{g}}_i\right) \\ \tilde{\boldsymbol{g}}_i = \mathrm{TopK}_k\left(\boldsymbol{g}_i\right) \end{cases} \tag{7}$$

where soft max$(\cdot)$ denotes the softmax function and $\mathrm{TopK}_k(\cdot)$ represents selecting the top k largest elements while setting the values of all other elements to $-\infty$. It can be seen from (6) and (7) that a larger $\tilde{g}_{ij}$ will correspond to a higher probability, indicating that the pixel will be selected and assigned a larger weight. Conversely, a smaller value will be set to $-\infty$, thereby becoming 0 after softmax activation, indicating that the pixel will be disregarded. Thus, k pixels containing more discriminative information can be selected from the h^2 pixels. The selected k pixels are then mixed according to their respective probabilities

$$\tilde{\boldsymbol{P}}^{(\mathrm{LiDAR})} = \sum_{j\in \mathrm{TopK}_k(\boldsymbol{g}_i)} p_{ij} \boldsymbol{P}_{ij}^{(\mathrm{LiDAR})} \tag{8}$$

where $\tilde{\boldsymbol{P}}^{(\mathrm{LiDAR})}$ represents the mixed PVs. By following (6)–(8), the number of PVs is significantly reduced, thereby lowering the computational cost for the subsequent basic blocks of the student model. Moreover, the discriminative pixels are effectively mixed according to their respective weights. The gating network is designed as a separate lightweight MLP and is not integrated into the basic blocks of the student FM. It operates externally to compute importance scores for LiDAR prompt selection, while all transformer layers of the student FM remain frozen.

C. Modified KD

The obtained HSI and LiDAR PVs are jointly fed into the student FM for feature extraction. It is worth noting that during the feature extraction process, all parameters of the basic blocks are frozen. Then, the multimodal features extracted by the basic blocks of the student model are

$$\boldsymbol{F}^{(\mathrm{S})} = f_{\boldsymbol{\theta}^{(\mathrm{S})}}\left(\left[\tilde{\boldsymbol{X}}^{(\mathrm{HSI})}, \boldsymbol{P}^{(\mathrm{spe})}, \tilde{\boldsymbol{P}}^{(\mathrm{LiDAR})}\right] + \boldsymbol{E}\right) \tag{9}$$

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where $\boldsymbol{\theta}^{(\mathrm{S})}$ denotes the parameters of the student model's basic blocks, $\boldsymbol{E}$ is the positional embeddings, and the detailed calculations are

$$\boldsymbol{F}_l^{(\mathrm{S})} = \mathrm{MHSA}\left(\mathrm{LN}\left(\left[\boldsymbol{F}_{l-1}^{(\mathrm{S})}, \boldsymbol{P}^{(\mathrm{spe})}, \tilde{\boldsymbol{P}}^{(\mathrm{LiDAR})}\right]\right)\right) \tag{10}$$

where $\mathrm{LN}(\cdot)$ represents the layer normalization. When $l = 1$, $\boldsymbol{F}_{l-1}^{(\mathrm{S})} = \tilde{\boldsymbol{X}}^{(\mathrm{HSI})}$. $\mathrm{MHSA}(\cdot)$ denotes the multihead self-attention, and the calculation of each head is

$$\mathrm{SA}\left(\boldsymbol{F}_l^{(\mathrm{S})}\right) = \mathrm{soft\,max}\left(\boldsymbol{Q}\boldsymbol{K}^{\mathrm{T}}/\sqrt{D'}\right)\boldsymbol{V} \tag{11}$$

where $\boldsymbol{Q} \in \mathbb{R}^{N\times D''}$, $\boldsymbol{K} \in \mathbb{R}^{N\times D''}$, and $\boldsymbol{V} \in \mathbb{R}^{N\times D''}$ are the queries, keys, and values of the basic blocks, respectively, and can be obtained by mapping $\boldsymbol{F}_l^{(\mathrm{S})}$ with three matrices. $N = N^{\mathrm{tr}} + N^{\mathrm{te}}$ represents the total number of labeled and unlabeled samples and $D'' = D/h$ denotes the dimension of embedding features for each head, where h is the number of heads and D is the original embedding dimension. Then, the output of the multihead self-attention can be obtained by combining the results of multiple single-head self-attention operations as follows:

$$\mathrm{MHSA}\left(\boldsymbol{F}_L^{(\mathrm{S})}\right) = \left[\mathrm{SA}_1\left(\boldsymbol{F}_L^{(\mathrm{S})}\right), \ldots, \mathrm{SA}_h\left(\boldsymbol{F}_L^{(\mathrm{S})}\right)\right]\boldsymbol{W} \tag{12}$$

where $\boldsymbol{W}$ is used to restore the structure to its original embedding dimension after concatenation along the channel dimension. Subsequently, a linear classifier is added to the student FM, and the predicted logits $\tilde{\boldsymbol{y}}^{(\mathrm{S})}$ on all samples can be obtained by

$$\tilde{\boldsymbol{y}}^{(\mathrm{S})} = f_{\boldsymbol{\theta}^{(\mathrm{C}),(\mathrm{S})}}\left(\boldsymbol{F}_L^{(\mathrm{S})}\right) \tag{13}$$

where $\boldsymbol{\theta}^{(\mathrm{C}),(\mathrm{S})}$ is the corresponding parameters of student model's classifier. For the predicted logits $\tilde{\boldsymbol{y}}$ corresponding to the labeled samples, we calculate the conventional supervised classification loss to measure the empirical loss of the student model

$$\mathcal{L}_{\mathrm{CE}}^{(\mathrm{S})} = -\frac{1}{N^{\mathrm{tr}}}\sum_{i=1}^{N^{\mathrm{tr}}} \boldsymbol{y}_i \log \tilde{\boldsymbol{y}}_i. \tag{14}$$

For the predicted logits of the unlabeled samples, KD can be reasonably considered. The original KD aims to transfer the knowledge from a heavy teacher model to a lightweight student model, so that the student model can master the expertise of the teacher model. This can be realized by aligning the output distribution of the two models. KD can also provide a candidate for implementing semi-supervised learning. Since the predicted logits have been calculated by the teacher model, we can treat them as the pseudo-supervisory information of unlabeled samples. Then, a KL divergence loss can be used to measure the difference between $\tilde{\boldsymbol{y}}_{\mathrm{TE}}^{(\mathrm{S})}$ and $\tilde{\boldsymbol{y}}_{\mathrm{TE}}$

$$\mathcal{L}_{\mathrm{KD}} = \mathrm{KL}\left(\tilde{\boldsymbol{y}}_{\mathrm{TE}} \,\|\, \tilde{\boldsymbol{y}}_{\mathrm{TE}}^{(\mathrm{S})}\right). \tag{15}$$

By doing so, a large amount of unlabeled samples can be utilized. However, the teacher model cannot be fully trusted, because it can still make incorrect predictions. Therefore, we propose a novel method to modify the KD procedure, as shown in Fig. 2.

For regular KD, the learning objective of the student model is to approximate the predicted logits of the teacher model (the red pentagram in Fig. 2). However, the logit predicted by the teacher model may be class-ambiguous. As shown in Fig. 2(a), the distances from the predicted logit to Classes 9 and 13 are close. Therefore, we reconstruct an extra logit by the predicted logits corresponding to the labeled samples (referred to as training logits) and use it as an additional objective to be approximated. Instead of using all the training logits, we select a more reliable subset from them. To achieve this goal, we first calculate the similarity between the unlabeled logits $\tilde{\boldsymbol{y}}_{i,\mathrm{TE}}$ and the training logits $\tilde{\boldsymbol{y}}_{\mathrm{TR}}$

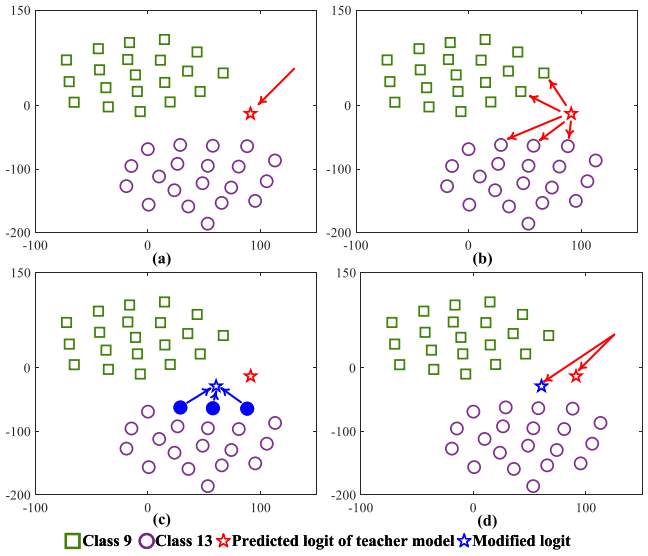


Fig. 2. Schematic of MKD: t-SNE visualization of training samples with Classes 9 and 13 on the Houston dataset. (a) Predicted teacher logit. (b) Similarity-based candidate logit selection. (c) Modified logit reconstruction. (d) Modified KD.

$$\mathcal{M}_i = \tilde{\boldsymbol{y}}_{i,\mathrm{TE}}^{\mathrm{T}} \times \tilde{\boldsymbol{y}}_{\mathrm{TR}} \tag{16}$$

where $\mathcal{M}_i \in \mathbb{R}^{N^{\mathrm{tr}}}$ represents the similarity between the ith test sample and all the training samples. Then, by sorting $\mathcal{M}_i$, we can obtain the M samples most similar to the test sample, forming a candidate subset $\boldsymbol{M}_i$

$$\boldsymbol{M}_i = \mathrm{TopK}_M\left(\mathcal{M}_i\right) \tag{17}$$

where $\mathrm{TopK}_M(\cdot)$ selects the M largest entries, and a candidate logits subset $\tilde{\boldsymbol{y}}_{i,\mathrm{TE}}$ can be obtained according to $\boldsymbol{M}_i$. In $\boldsymbol{M}_i$, all the training logits have high similarity with the unlabeled logit, but they may come from multiple classes. Therefore, we count the occurrence of each class in $\tilde{\boldsymbol{y}}_{i,\mathrm{TE}}$ to obtain $n_C = \{\mathrm{count}(1), \ldots, \mathrm{count}(C)\}$, where $\mathrm{count}(i) = \sum_{y_{j,\mathrm{TR}}\in\boldsymbol{M}_i} \mathbb{I}(y_j = i)$. Then, we identify the top c occurring classes $\boldsymbol{y}_c = \{y_1, \cdots y_c\}$ and select the corresponding training logits in $\tilde{\boldsymbol{y}}_{i,\mathrm{TE}}$ to form the final candidate set $\boldsymbol{m}_i \in \mathbb{R}^{m\times C}$

$$\boldsymbol{m}_i = \left\{\tilde{\boldsymbol{y}}_{j,\mathrm{TR}} | \tilde{\boldsymbol{y}}_{j,\mathrm{TR}} \in \boldsymbol{M}_i, y_{j,\mathrm{TR}} \in \boldsymbol{y}_c\right\}. \tag{18}$$

After the above process, the selected training logits in $\boldsymbol{m}_i$ are more reliable. Then, the reconstructed logit $\boldsymbol{y}_{i,\mathrm{TE}}^{(\mathrm{R})}$ can be obtained by

$$\begin{cases} \boldsymbol{S}_i = \mathrm{soft\,max}\left(\tilde{\boldsymbol{y}}_{i,\mathrm{TE}}^{\mathrm{T}}\boldsymbol{m}_i\right) \\ \boldsymbol{y}_{i,\mathrm{TE}}^{(\mathrm{R})} = \sum_{j=1}^{m} \boldsymbol{S}_{ij} m_{ij} \end{cases} \tag{19}$$

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where $\boldsymbol{S}_i \in \mathbb{R}^m$ represents the correlation between m_i and the logits of the ith unlabeled logit. Since the labels of all training logits in $\boldsymbol{m}_i$ are known, using these logits to reconstruct the modified logits $\boldsymbol{y}_{i,\mathrm{TE}}^{(\mathrm{R})}$ results in stronger discriminative ability. Furthermore, we assume that some classes are correlated with the ground-truth label, while the others are not. Therefore, similar to (19), we also reconstruct a negative logit for each test sample by

$$\begin{cases} \hat{\boldsymbol{S}}_i = \text{soft max}\left(-\tilde{\boldsymbol{y}}_{i,\mathrm{TE}}^{\mathrm{T}} \boldsymbol{m}_i\right) \\ \hat{\boldsymbol{y}}_{i,\mathrm{TE}}^{(\mathrm{R})} = \sum_{j=1}^{m} \hat{\boldsymbol{S}}_{ij} m_{ij}. \end{cases} \tag{20}$$

Then, the larger the value in $\hat{\boldsymbol{y}}_{i,\mathrm{TE}}^{(\mathrm{R})}$, the lower the relevance of the class to the ground-truth label. In this article, we distribute the predictive probability of the least relevant class evenly among the other classes to obtain $\tilde{\boldsymbol{y}}_{i,\mathrm{TE}}^{(\mathrm{R})}$, thereby reducing the class ambiguity of the learned model. Finally, $\tilde{\boldsymbol{y}}_{i,\mathrm{TE}}^{(\mathrm{R})}$ can be considered as another target for the student to approximate

$$\mathcal{L}_{\mathrm{MKD}} = \mathrm{KL}\left(\tilde{\boldsymbol{y}}_{i,\mathrm{TE}}^{(\mathrm{R})} \,\|\, \tilde{\boldsymbol{y}}_{i,\mathrm{TE}}^{(\mathrm{S})}\right). \tag{21}$$

MPT-MKD can implicitly alleviate the class-imbalance-induced bias in pseudo-supervision information over unlabeled samples. This is because the modified soft target is reconstructed from labeled sample logits rather than from the whole unlabeled set, as labeled samples can provide class anchors for reconstructing the modified logits. Moreover, the Top-K similarity search and top-c class-consistency filtering are performed at the sample level instead of according to the global pseudo-label frequency. Therefore, when an unlabeled minority-class sample is close to minority-class anchors in the teacher-logit space, these anchors can be used to modify the original teacher prediction, reducing the possibility that majority-class pseudo-labels dominate the distillation process.

D. Overall Objective

In summary, the overall objective $\mathcal{L}$ of the student model includes the supervised loss $\mathcal{L}_{\mathrm{CE}}^{(\mathrm{S})}$, the KD loss $\mathcal{L}_{\mathrm{KD}}$, and the MKD loss $\mathcal{L}_{\mathrm{MKD}}$

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}}^{(\mathrm{S})} + \mathcal{L}_{\mathrm{KD}} + \alpha \mathcal{L}_{\mathrm{MKD}} \tag{22}$$

where α is the balancing coefficient. Then, all parameters in the basic blocks of the student FM are frozen, while the learnable parameters, including $\boldsymbol{\theta}^{(\mathrm{SPL})}$, $\boldsymbol{\theta}^{(\mathrm{2D})}$, and $\boldsymbol{\theta}^{(\mathrm{C})}$, are optimized. The optimal values of these learnable parameters can be obtained using gradient-descent-based algorithms, such as Adam. The overall pipeline of MPT-MKD is shown in Algorithm 1.

Algorithm 1 Overall Pipeline of MPT-MKD

Input: Original HSI $\boldsymbol{X}^{(\mathrm{HSI})}$ and LiDAR data $\boldsymbol{X}^{(\mathrm{LiDAR})}$, the teacher FM $f_{\boldsymbol{\theta}^{(\mathrm{T})}}$, the student FM $f_{\boldsymbol{\theta}^{(\mathrm{S})}}$, linear classifier $f_{\boldsymbol{\theta}^{(\mathrm{C})}}$ and $f_{\boldsymbol{\theta}^{(\mathrm{C}),(\mathrm{S})}}$, spectral prompt learner $f_{\boldsymbol{\theta}^{(\mathrm{SPL})}}$, LiDAR prompt learner $f_{\boldsymbol{\theta}^{(\mathrm{LiDAR})}}$, gating network $G_{\boldsymbol{\theta}^{(\mathrm{Gate})}}$, positional embeddings $\boldsymbol{E}$, and hyperparameters α, M.

Output: Classification results

Initialization: The teacher and student FMs are initialized with pretrained parameters, while the linear classifiers, spectral prompt learner, LiDAR prompt learner, gating network, and the positional embeddings are randomly initialized.

Step 1: Obtain the local neighborhood representations $\tilde{\boldsymbol{X}}^{(\mathrm{HSI})}$ and $\tilde{\boldsymbol{X}}^{(\mathrm{LiDAR})}$ for the HSI and LiDAR data, respectively.

Step 2: A learnable linear classifier is appended to the head of the teacher FM, and its optimal parameters are obtained using (2). Subsequently, the predicted logits of a test sample are computed using (3).

Step 3: Obtain the spectral PV and the LiDAR PV using (4) and (5).

Step 4: The mixed LiDAR PV $\tilde{\boldsymbol{P}}^{(\mathrm{LiDAR})}$ is obtained using (7) and (8) and then concatenated with the spectral PVs before being fed into the attention module of the student FM.

Step 5: Calculate the reconstructed logits and negative logits using (19) and (20), respectively.

Step 6: The cross-entropy loss, KD loss, and MKD loss are jointly computed on both the labeled and unlabeled datasets, as described in (22).

Step 7: The learnable parameters, including $\boldsymbol{\theta}^{(\mathrm{SPL})}$, $\boldsymbol{\theta}^{(\mathrm{2D})}$, and $\boldsymbol{\theta}^{(\mathrm{C})}$ are updated, and together with the frozen parameters of the student FM, they are used to obtain the final classification results.

III. Experimental Results and Analysis

To verify the effectiveness of the proposed method, comparative experiments were conducted on three real HSI and LiDAR datasets, including Houston 2013 [39], MUUFL [40], and Augsburg [41]. All the three datasets contain HSI and LiDAR data. For the Houston 2013 dataset, the HSI includes 144 bands, the LiDAR data contain 1 channel, and there are 15 different land cover classes. For the MUUFL dataset, it contains a total of 64 bands, and the LiDAR data include 2 channels with 11 different land cover classes. The HSI and LiDAR data contained in the Augsburg dataset include 180 spectral bands and 1 channel, respectively. It encompasses seven different land cover classes. More sample information is available in Figs. 3–5 and Table I.

A. Comparative Experiments

The following methods were selected for the comparative experiments: DSHFNet [4], NNCNet [18], HCT [6], ExViT [22], M2FNet [27], MICF-Net [28], MCFNet [15], and SPT [36]. To reduce the impact of random factors, all experiments were repeated five times, and the results are reported as the mean $\pm$ standard deviation. The hyperparameters of the corresponding comparative methods were selected according to the respective References. All the experiments are performed on a computer equipped with 3.70-GHz Intel Core i9-10900KF CPU, 32 GB of RAM, and RTX 4090 GPU, and the experimental environment is PyTorch. The network structures of the spectral prompt learner and LiDAR prompt learner are shown in Table II. The Adam optimizer is selected and the learning rate is set to 2×10^{-5}. The batch sizes of labeled samples and unlabeled samples are set to 16 and 320, respectively. The huge and base versions of

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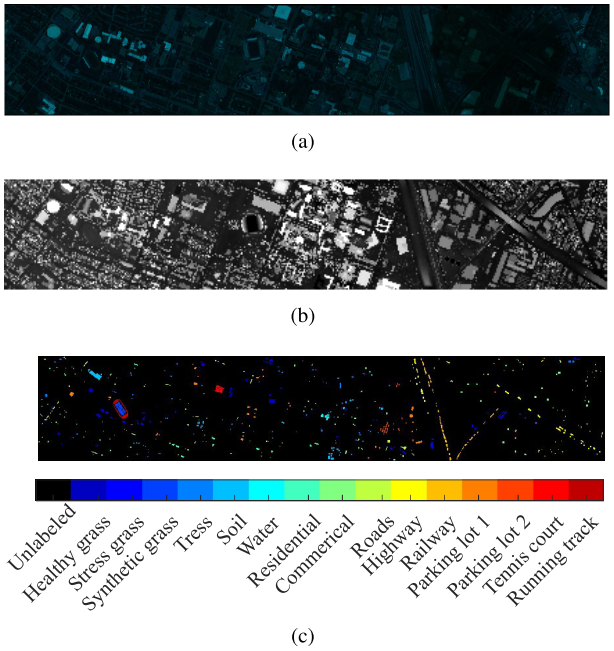


Fig. 3. Houston 2013 dataset. (a) Pseudo-color composite image of HSI. (b) LiDAR DSM. (c) Ground-truth map.

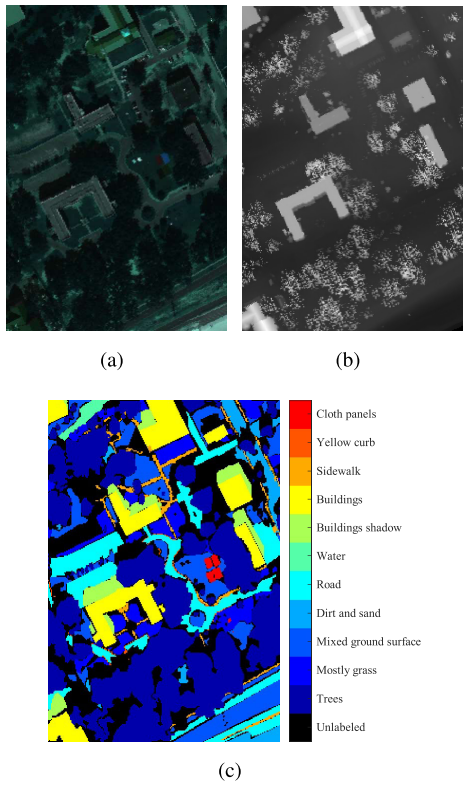


Fig. 4. MUUFL dataset. (a) Pseudo-color composite image of HSI. (b) LiDAR DSM. (c) Ground-truth map.

SpatSIGMA are selected as the teacher and student models, respectively.

The classification results of different methods on the three datasets are shown in Tables III–V and Figs. 6–11, and it can be seen that the following conditions hold.

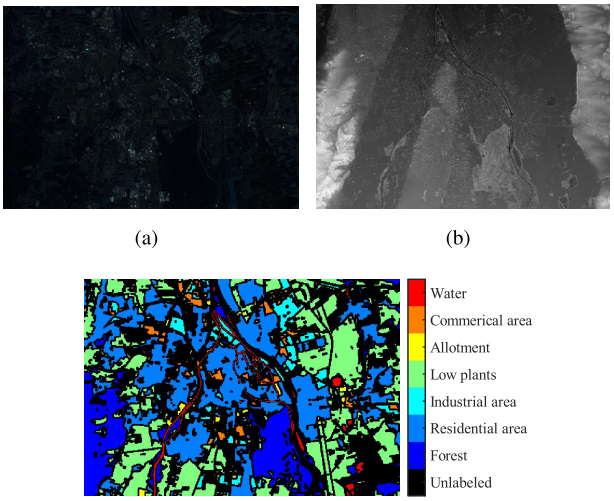


Fig. 5. Augsburg dataset. (a) Pseudo-color composite image of HSI. (b) LiDAR DSM. (c) Ground-truth map.

1) MPT-MKD can achieve the highest overall accuracy (OA) on the three datasets for the following reasons. First, the SpatSIGMA is an FM with a massive number of parameters, which makes it highly capable of extracting general spectral features. Second, by incorporating PVs of HSI and LiDAR data and applying the self-attention mechanism in the FM's basic blocks, the extracted features can simultaneously reflect both spectral and elevation information of ground objects. Finally, the MKD can ensure that the student model receives sufficient and relatively accurate supervisory information for training.
2) Compared to the Houston 2013 dataset, all methods showed relatively lower performance on the MUUFL and Augsburg datasets. This can be attributed to two main factors. First, the limited number of labeled samples makes it challenging to accurately capture the underlying data distribution, leading to overfitting in DL models and subsequently reducing classification performance. Second, the MUUFL and Augsburg datasets exhibit a higher degree of interclass similarity. For instance, on the MUUFL dataset, the "Road" and "Mixed ground surface" classes share considerable spatial and spectral similarities, resulting in significant misclassifications on these classes for all methods. Nonetheless, MPT-MKD consistently achieved the highest OA among all methods, demonstrating its superiority.
3) MPT-MKD achieved the highest or nearly highest classification accuracy on most classes. This is because MPT-MKD not only leverages the FM to extract discriminative spectral features from HSI but also introduces LiDAR PVs, utilizing a self-attention mechanism of the FM's basic blocks to enable the interaction between different modalities, thereby achieving effective feature fusion of HSI and LiDAR data.
4) In the local region of the Augsburg dataset, all methods suffer from varying degrees of misclassification. Among them, the MPT-MKD achieved the smallest number

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TABLE I
SAMPLE INFORMATION FOR THE THREE DATASETS

	Houston 2013			MUUFL			Augsburg		
	Class Name	Labeled	Unlabeled	Class Name	Labeled	Unlabeled	Class Name	Labeled	Unlabeled
C1	Healthy Grass	20	1231	Trees	20	23226	Forest	20	13487
C2	Stressed Grass	20	1234	Mostly Grass	20	4250	Residential Area	20	30309
C3	Synthetic Grass	20	677	Mixed Ground Surface	20	6862	Industrial Area	20	3831
C4	Trees	20	1224	Dirt and Sand	20	1806	Low Plants	20	26837
C5	Soil	20	1222	Road	20	6667	Allotment	20	555
C6	Water	20	305	Water	20	446	Commercial Area	20	1625
C7	Residential	20	1248	Buildings Shadow	20	2213	Water	20	1510
C8	Commercial	20	1224	Buildings	20	6220			
C9	Road	20	1232	Sidewalk	20	1365			
C10	Highway	20	1207	Yellow Curb	20	163			
C11	Railway	20	1215	Cloth Panels	20	249			
C12	Parking Lot 1	20	1213						
C13	Parking Lot 2	20	449						
C14	Tennis Court	20	408						
C15	Running Track	20	640						
	Total	300	14729		220	53467		140	78154

TABLE II
STRUCTURE OF SPECTRAL AND LIDAR PROMPT LEARNER

	Dataset	Input size	Layer type	Kernel number @ size	Step size/padding size	Output size
Spectral prompt learner	Houston 2013	144	1D Conv.	768 @ 36	36/0	768×4
	MUUFL	64	1D Conv.	768 @ 16	16/0	768×4
	Augsburg	180	1D Conv.	768 @ 45	45/0	768×4
LiDAR prompt learner	Houston 2013	11×11×1	2D Conv.	768 @ 3×3	1/1	11×11×768
	MUUFL	11×11×2	2D Conv.	768 @ 3×3	1/1	11×11×768
	Augsburg	11×11×1	2D Conv.	768 @ 3×3	1/1	11×11×768

TABLE III
COMPARISON OF CLASSIFICATION PERFORMANCE (HOUSTON 2013, %)

	DSHFNet [4]	MNCNet [18]	IJCAI [6]	FusAtNet [22]	M2FNet [27]	MICF-Net [28]	MACNet [15]	SPT [36]	MPT-MKD
C1	90.48±2.34	94.17±2.55	**96.18±1.50**	91.50±3.33	92.35±3.20	95.05±6.63	92.28±1.38	95.68±1.61	94.31±2.77
C2	96.29±6.57	95.93±2.00	**98.38±1.23**	97.88±0.94	93.11±2.51	89.30±6.60	93.06±0.53	97.31±1.69	94.42±4.83
C3	99.26±0.92	99.88±0.13	99.20±0.54	98.53±1.25	98.49±1.10	**100.00±0.00**	99.55±0.15	99.94±0.13	99.91±0.20
C4	96.67±3.05	96.29±2.90	98.19±2.65	97.91±1.33	95.05±2.11	89.55±3.50	**98.52±1.23**	98.27±2.76	97.91±2.27
C5	99.51±0.59	99.89±0.21	99.98±0.04	99.61±0.35	96.75±1.63	95.31±4.97	**100.00±0.00**	99.84±0.28	**100.00±0.00**
C6	91.7-±4.75	98.76±1.76	95.74±6.24	94.29±5.90	94.23±5.24	97.96±1.41	**98.85±1.15**	95.93±6.20	94.69±5.75
C7	93.0-±2.60	91.65±3.09	91.73±1.36	**95.77±1.68**	93.59±0.97	92.06±2.65	94.38±3.61	95.11±2.28	94.44±1.41
C8	76.91±4.64	71.67±-.76	83.04±5.77	79.36±5.82	83.20±7.14	**94.49±4.39**	80.51±9.27	75.95±4.66	80.83±6.69
C9	56.22±9.17	85.84±0.60	80.96±4.60	87.70±7.88	86.95±2.58	**90.73±4.25**	90.70±3.21	87.86±3.84	89.53±3.96
C10	90.21±6.80	94.78±2.98	93.16±9.19	91.35±4.87	85.56±5.67	80.55±9.73	86.53±-.77	88.45±3.45	**95.00±3.71**
C11	93.25±3.49	97.47±3.50	97.78±3.12	97.42±1.02	95.59±2.00	86.40±5.40	**98.06±0.62**	92.07±2.89	94.26±5.04
C12	**95.26±2.81**	91.89±4.22	94.74±5.82	92.93±2.42	87.85±4.08	86.05±5.91	92.53±1.28	91.82±1.21	94.58±2.72
C13	92.74±1.48	97.51±2.49	96.62±1.74	96.44±1.50	94.34±2.03	75.89±6.50	**98.99±0.11**	94.61±1.05	95.46±1.85
C14	99.51±1.10	**100.00±0.00**	99.95±0.11	99.66±0.54	98.09±1.80	89.76±6.24	**100.00±0.00**	99.75±0.42	**100.00±0.00**
C15	**100.00+0.00**	99.97+0.07	**100.00−0.00**	99.88+0.28	99.50−1.00	97.13+1.69	**100.00−0.00**	**100.00+0.00**	**100.00+0.00**
AA	91.-0±0.41	94.38±0.33	**95.04±0.30**	94.68±0.64	92.98±0.44	90.68±0.60	94.93±0.52	94.37±0.47	95.02±0.60
OA	90.86±0.87	93.43±0.72	94.28±1.31	93.99±0.88	92.09±0.36	89.85±1.04	93.83±0.57	93.54±0.52	**94.36±0.60**
Kappa	88.40±1.45	92.90±0.77	93.79±1.40	93.50±0.95	91.45±0.40	89.03±1.13	93.32±0.62	93.02±0.56	**93.90±0.65**
Time (s)	1537	849	1301	227	281	1381	625	**96**	2235

of misclassified pixels. A similar phenomenon is also observed in the local regions on the Houston 2013 and MUUFL datasets. Moreover, compared with the other methods, MPT-MKD can produce smoother and more detailed classification maps. Other methods are more likely to misclassify classes on the MUUFL dataset, such as confusing "Trees" and "Dirt and sand" with "Cloth panels," or mislabeling "Water" and "Residential area" as "Low plants."

5) MPT-MKD requires more computational time than most competing methods. This is mainly due to two factors. First, our framework adopts a teacher–student distillation paradigm and leverages a large amount of unlabeled samples, which introduces additional forward

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TABLE IV
COMPARISON OF CLASSIFICATION PERFORMANCE (MUUFL, %)

	DSHFNet [4]	NNCNet [18]	HCT [6]	ExViT [22]	M2FNet [27]	MICF-Net [28]	MCFNet [15]	SPT [36]	MPT-MKD
C1	77.78±8.29	79.39±3.67	83.16±2.74	85.81±1.32	87.55±1.00	**98.08±1.04**	88.29±2.71	83.26±1.01	89.02±3.61
C2	77.62±9.58	82.71±3.77	74.42±5.67	77.56±6.63	77.08±5.20	62.16±5.91	79.87±5.68	**84.56±2.37**	78.87±7.36
C3	72.81±7.69	52.28±15.70	64.09±5.05	59.61±3.25	50.76±6.74	**80.85±7.25**	58.77±8.41	65.31±5.48	66.97±7.47
C4	91.34±4.83	85.33±5.83	87.23±8.00	91.25±1.75	80.68±8.46	79.08±8.53	**93.33±3.31**	86.62±4.48	86.18±2.29
C5	75.96±7.77	70.94±9.98	74.77±4.95	78.76±4.76	75.59±7.04	**93.90±2.00**	85.17±3.41	88.76±5.68	86.05±2.33
C6	99.73±0.60	98.25±1.87	99.87±0.20	99.42±1.18	98.88±1.44	69.94±9.17	**100±0.00**	99.37±1.17	89.78±4.51
C7	**93.35±4.10**	75.33±8.09	85.20±6.98	82.71±9.69	83.16±8.43	64.42±9.35	83.34±4.99	89.30±5.89	83.59±4.35
C8	87.66±5.34	89.79±2.02	91.07±1.67	92.83±2.73	88.94±2.69	**95.77±2.08**	95.23±1.24	88.04±3.40	81.40±2.48
C9	44.85±15.03	53.45±6.22	51.68±7.87	51.66±6.09	53.41±4.05	42.30±5.24	53.75±10.56	**74.99±3.93**	59.97±7.33
C10	42.33±17.42	75.09±10.72	82.08±5.69	71.17±3.42	74.85±5.26	6.33±1.41	59.51±27.52	**94.11±5.20**	91.53±4.45
C11	95.74±0.36	**97.99±1.63**	96.38±2.02	95.98±0.00	95.58±0.57	66.30±9.35	95.96±0.14	94.30±1.48	95.58±1.73
AA	78.11±2.77	78.23±2.63	80.90±1.24	80.62±1.40	78.77±2.34	69.01±1.16	80.83±3.11	**86.24±1.45**	82.63±0.71
OA	78.47±3.67	76.00±1.31	79.50±1.71	81.09±1.22	79.47±1.98	82.58±1.39	83.06±1.12	82.67±1.51	**83.11±1.54**
Kappa	72.98±4.18	69.36±1.89	73.88±2.02	75.76±1.53	73.69±2.46	77.69±1.64	**78.48±1.48**	77.91±1.89	77.95±1.81
Time	575	460	**24**	201	468	2081	808	272	2470

TABLE V
COMPARISON OF CLASSIFICATION PERFORMANCE (AUGSBURG, %)

	DSHFNet [4]	NNCNet [18]	HCT [6]	ExViT [22]	M2FNet [27]	MICF-Net [28]	MCFNet [15]	SPT [36]	MPT-MKD
C1	84.79±11.91	93.95±1.62	94.67±1.74	93.63±1.31	93.95±1.62	93.21±4.75	90.91±5.18	**96.18±2.13**	94.49±2.45
C2	65.34±13.33	76.55±7.48	80.90±5.26	80.29±3.54	76.55±7.48	93.17±4.21	**93.47±4.33**	81.04±3.66	89.82±5.35
C3	35.84±21.34	61.40±3.53	62.53±18.00	53.67±14.27	61.40±3.53	52.81±5.30	49.01±0.99	**62.61±4.73**	51.32±8.97
C4	81.30±10.13	74.48±6.98	87.48±7.45	74.32±3.53	74.48±6.98	95.09±3.93	**95.76±1.71**	80.77±5.82	89.38±7.00
C5	86.38±10.77	85.44±3.32	89.22±3.29	81.62±4.35	85.44±3.32	8.88±7.91	16.02±6.22	88.18±7.30	**96.47±2.07**
C6	**66.82±21.66**	59.24±5.05	51.43±8.74	62.46±6.77	59.24±5.05	25.69±8.49	18.96±0.49	37.61±8.56	53.21±11.66
C7	70.25±5.08	65.32±5.26	**71.75±6.76**	70.40±2.14	65.32±5.26	50.68±14.18	56.48±6.59	65.68±6.10	67.02±1.74
AA	70.10±3.63	73.77±2.61	76.85±4.61	73.77±1.68	73.77±2.61	59.93±2.55	60.09±0.78	73.15±1.37	**77.39±1.35**
OA	73.01±7.56	77.63±5.26	83.91±5.03	78.68±2.62	77.63±5.26	79.44±3.99	83.34±0.05	81.50±2.79	**87.43±1.77**
Kappa	64.47±8.45	69.86±6.40	78.12±6.51	71.06±3.28	69.86±6.40	69.36±8.10	77.03±0.29	74.64±3.50	**82.34±2.52**
Time	834	618	**165**	195	618	1036	1033	389	1605

TABLE VI
NUMBER OF LEARNABLE AND FROZEN PARAMETERS IN DIFFERENT METHODS (M)

	Houston 2013		MUUFL		Augsburg	
	Learnable	Frozen	Learnable	Frozen	Learnable	Frozen
DSHFNet [4]	0.54	0.00	0.51	0.00	0.55	0.00
NNCNet [18]	2.75	0.00	2.70	0.00	2.74	0.00
HCT [6]	0.45	0.00	0.46	0.00	0.45	0.00
ExViT [22]	0.24	0.00	0.23	0.00	0.24	0.00
M2FNet [27]	0.20	0.00	0.16	0.00	0.23	0.00
MICF-Net [28]	0.18	0.00	0.14	0.00	0.20	0.00
MCFNet [15]	0.54	0.00	0.47	0.00	0.66	0.00
SPT [36]	0.05	151.28	0.04	151.28	0.06	151.28
MPT-MKD	1.50	237.13	1.37	237.13	1.36	237.13

passes and similarity computations for constructing the modified logits. Second, both the teacher and student are built upon a large-scale FM, whose computational cost is inherently high. Despite the increased consumed time, the proposed method consistently achieves higher OA, indicating that the additional computational cost effectively translates into improved classification performance.

6) The labeled set is class-balanced, but the unlabeled set exhibits a natural class imbalance. Despite this imbalance, MPT-MKD does not simply favor the majority classes. For example, on the Houston 2013 dataset, the minority classes "Water," "Parking Lot 2," "Tennis Court," and "Running Track" contain only 305, 449, 408, and 640 unlabeled samples, respectively, while MPT-MKD obtains 94.69%, 95.46%, 100%, and 100% on these classes. On the MUUFL dataset, the two smallest classes, "Yellow Curb" and "Cloth Panels," contain only 163 and 249 unlabeled samples, while MPT-MKD achieves 91.53% and 95.58%, respectively. On the Augsburg dataset, the minority class "Allotment" has only 555 unlabeled samples, and

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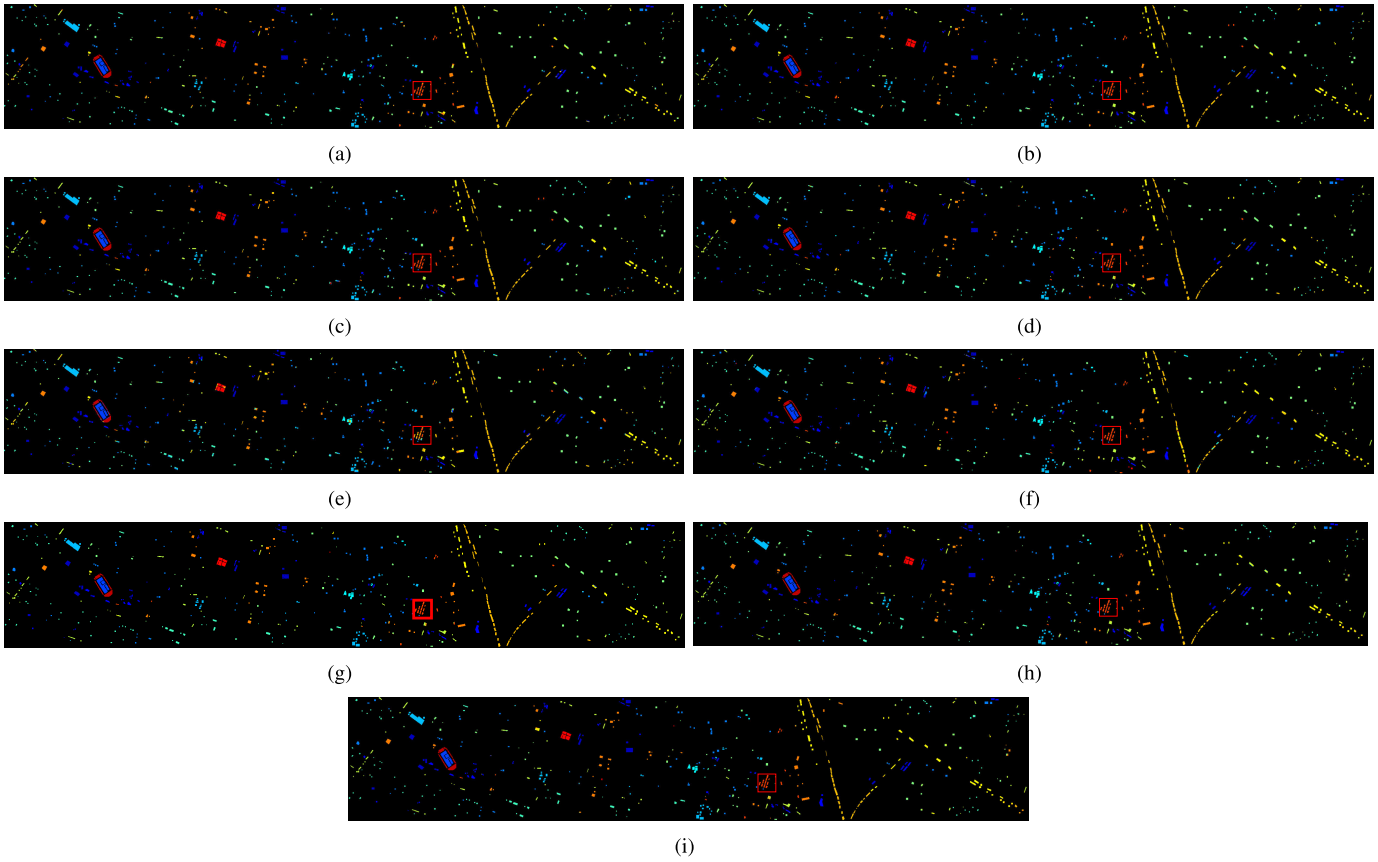


Fig. 6. Classification maps obtained by different methods on the Houston 2013 dataset. (a) DSHFNet. (b) NNCNet. (c) HCT. (d) ExViT. (e) M2FNet. (f) MICF-Net. (g) MCFNet. (h) SPT. (i) MPT-MKD.

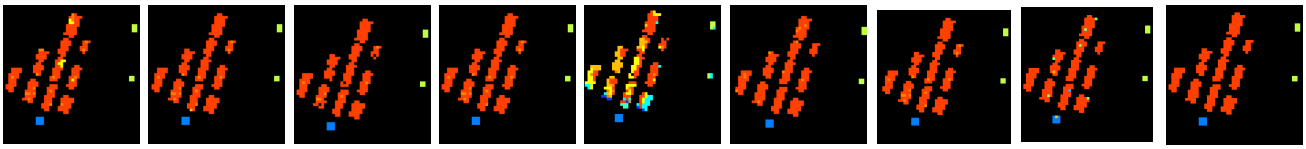


Fig. 7. Classification maps of a local region obtained by different methods on the Houston 2013 dataset. (a) DSHFNet. (b) NNCNet. (c) HCT. (d) ExViT. (e) M2FNet. (f) MICF-Net. (g) MCFNet. (h) SPT. (i) MPT-MKD.

MPT-MKD reaches the highest accuracy of 96.47%. These results indicate that the modified logits reconstructed from class-balanced labeled anchors help the student model preserve supervision for rare classes and reduce the risk that majority-class pseudo-labels dominate the KD process.

Besides, to evaluate the model complexity of the proposed method, the numbers of learnable and frozen parameters of different methods are shown in Table VI. First, the number of trainable parameters in MPT-MKD is neither the highest nor the lowest, but falls within an acceptable range. Then, MPT-MKD has more trainable parameters than SPT, a pretraining-based method, as it includes parameters from both teacher and student models. Moreover, compared to full fine-tuning, MPT-MKD requires fine-tuning only about 0.63% of the total model parameters, demonstrating its high parameter efficiency.

B. Parameter Analysis

To evaluate the impact of hyperparameters of MPT-MKD on the classification performance, a series of parameter analysis experiments was conducted. First, the effect of the balance factor α on the OA is analyzed, as shown in Fig. 12. α governs the tradeoff between teacher guidance (KD) and ambiguity-reduced corrective supervision (MKD). On the three datasets, as the value of α increases, the OA generally exhibits a trend of first increasing and then decreasing. On the one hand, a small value of α, such as the extreme case of $\alpha = 0$, implies that no modified logits are included in the learning process. In this scenario, the student model becomes overly reliant on the pseudo-supervisory information from the teacher model. If the teacher model generates incorrect predictions, it can mislead the student model, thereby reducing its discriminative capability. On the other hand, if α is set too large, the learning process may become overly dependent on the reconstructed logits, which are derived from a limited labeled subset and may overemphasize local neighborhood structure while weakening both the stabilizing effect of teacher guidance and the direct supervised signal. Therefore, OA typically peaks at a moderate α, where teacher supervision and correction complement each other. Experimental results indicate that setting $\alpha = 0.4$ achieves the highest or nearly highest OA on all three datasets.

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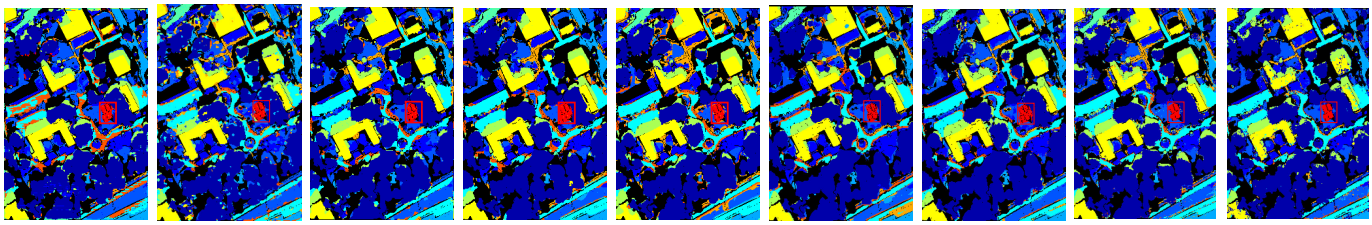


Fig. 8. Classification maps obtained by different methods on the MUUFL dataset. (a) DSHFNet. (b) NNCNet. (c) HCT. (d) ExViT. (e) M2FNet. (f) MICF-Net. (g) MCFNet. (h) SPT. (i) MPT-MKD.

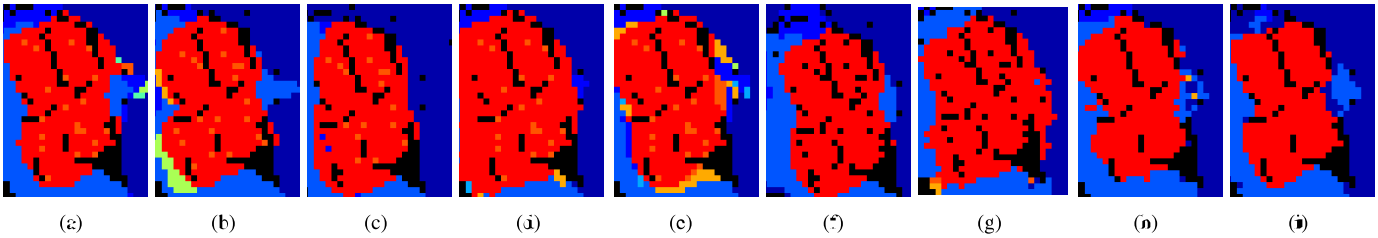


Fig. 9. Classification maps of a local region obtained by different methods on the MUUFL dataset. (a) DSHFNet. (b) NNCNet. (c) HCT. (d) ExViT. (e) M2FNet. (f) MICF-Net. (g) MCFNet. (h) SPT. (i) MPT-MKD.

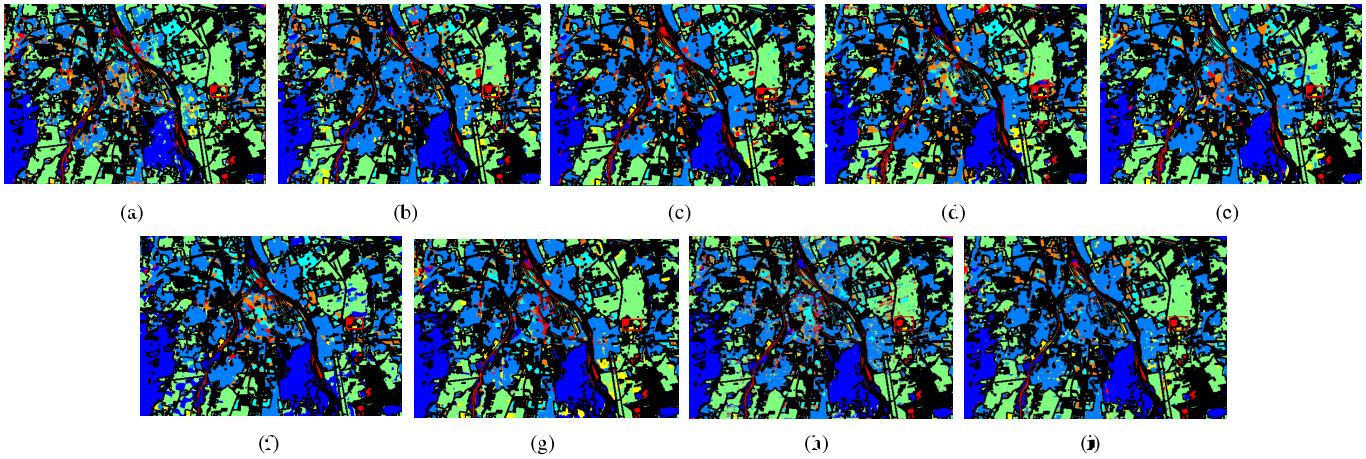


Fig. 10. Classification maps obtained by different methods on the Augsburg dataset. (a) DSHFNet. (b) NNCNet. (c) HCT. (d) ExViT. (e) M2FNet. (f) MICF-Net. (g) MCFNet. (h) SPT. (i) MPT-MKD.

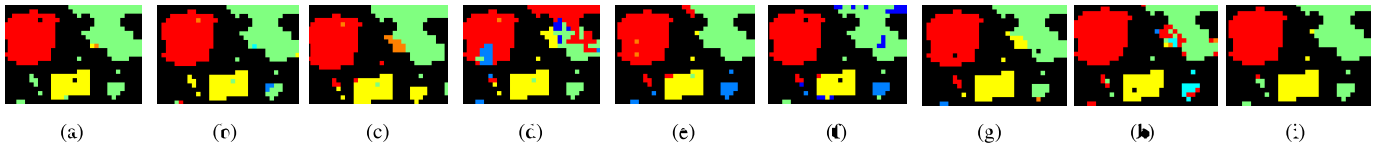


Fig. 11. Classification maps of a local region obtained by different methods on the Augsburg dataset. (a) DSHFNet. (b) NNCNet. (c) HCT. (d) ExViT. (e) M2FNet. (f) MICF-Net. (g) MCFNet. (h) SPT. (i) MPT-MKD.

Therefore, $\alpha = 0.4$ is recommended for optimizing classification performance.

Next, we analyze the impact of the number of labeled samples M in the candidate set on the OA, as shown in Fig. 13. As M increases, the OA remains relatively stable. When M is sufficiently large to cover the relevant neighboring classes, the subsequent top-c filtering makes the reconstructed logits insensitive to further increases in M. Empirically, M can be selected as 1.5–3 times the number of labeled samples per class. On all three datasets, when M exceeds 30, nearly all samples corresponding to the target classes are included in the candidate set and utilized for reconstructing the correction logits. Therefore, $M = 30$ is recommended.

Subsequently, the impact of varying numbers of labeled samples on the OA was further investigated, as shown in Fig. 14. It can be seen from the figure that the OA of all methods generally increases with the number of labeled samples. This is because a larger number of labeled samples provide richer supervisory information, thereby enabling a more comprehensive characterization of the true probability

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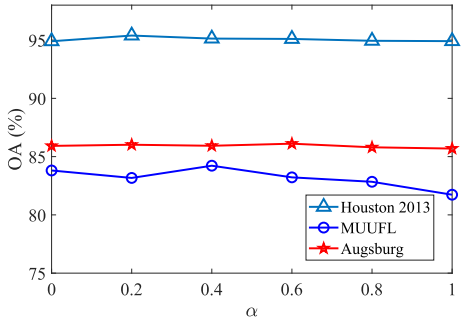


Fig. 12. Relationship between α and OA on the three datasets.

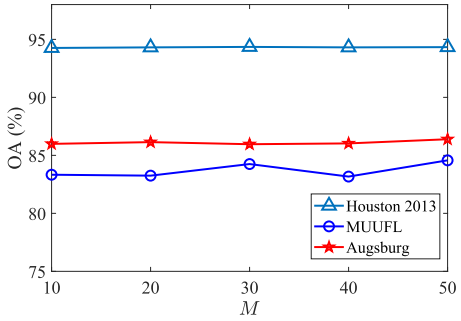


Fig. 13. Relationship between M and OA on the three datasets.

distribution. Among all the methods, MPT-MKD achieves the highest OA in most cases, which can be attributed, on the one hand, to the strong feature extraction capability of the spectral FM and, on the other hand, to the effectiveness of the proposed fine-tuning strategy.

C. Ablation Study and Analysis

To evaluate the impact of different components on the classification performance of MPT-MKD, an ablation study was conducted, as shown in Fig. 15. In this study, the following configurations were analyzed.

1) *SpatSIGMA:* This baseline model uses a limited set of labeled samples for training. The parameters within the SpatSIGMA's basic blocks remain frozen, while only the parameters of the additional classifier are updated.
2) *SpatSIGMA w/ PT:* This configuration incorporates the PT into the SpatSIGMA.
3) *SpatSIGMA w/ MKD:* This configuration combines the SpatSIGMA with the MKD strategy to assess the effect of transferring knowledge from a teacher model.
4) *MPT-MKD w/o MKD, MPT-MKD w/o Spectral PT, and MPT-MKD w/o LiDAR PT:* This configuration denotes the componentwise ablations to verify their contributions.

As shown in Fig. 15, MPT-MKD consistently achieves the highest OA on all three datasets. This superior performance is attributed to two main factors. On the one hand, the extensive supervised information provided by the teacher model ensures that the additional learnable parameters are sufficiently trained. The modified mechanism further enhances the reliability of the supervised information, resulting in higher OA; on the other hand, the integration of elevation

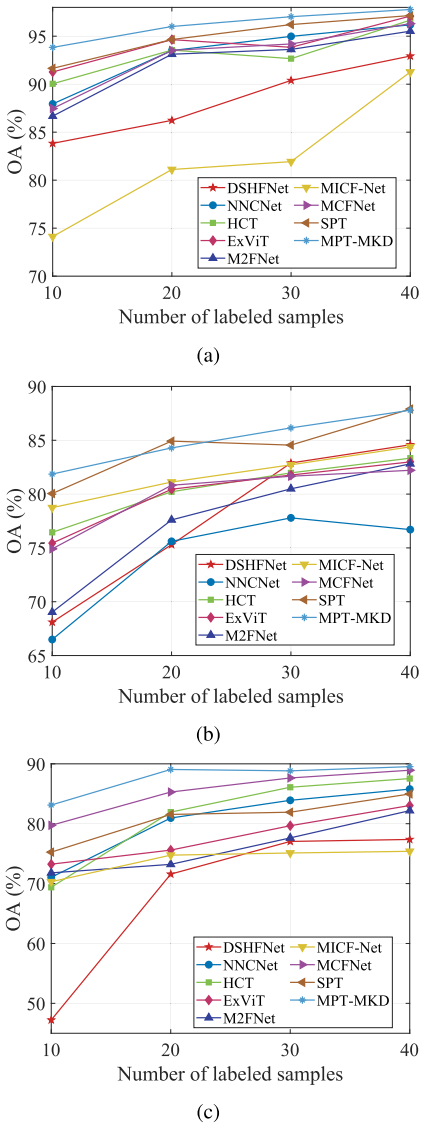


Fig. 14. Impact of labeled sample size on OA. (a) Houston 2013. (b) MUUFL. (c) Augsburg.

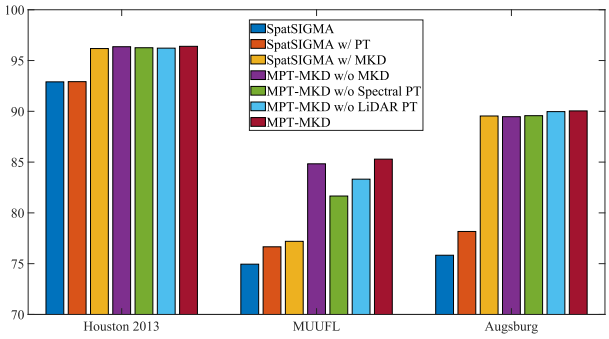


Fig. 15. Ablation experiments.

information through multimodal prompts enables the model to capture more discriminative features, thereby improving its classification performance. Furthermore, the componentwise ablation experiments demonstrate the impact of each individual component on OA, indicating that the integrated use of

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KD, MPT, and the modified mechanism plays a crucial role in maximizing classification accuracy.

Removing spectral PT leads to a larger performance degradation on the MUUFL dataset. This is mainly because the MUUFL dataset contains several low-height and material-sensitive classes, such as "Road" and "Sidewalk," which exhibit weak elevation contrasts and therefore rely more heavily on fine-grained spectral features for discrimination. Spectral PT provides the key trainable path for adapting the frozen HSI representation. Without spectral PT, the student model mainly depends on frozen spectral tokens, LiDAR prompts, and the classifier, which weakens spectral adaptation and increases confusion among classes with subtle spectral differences.

IV. Conclusion

To generalize the remote sensing FM (SpatSIGMA) to the joint classification task of HSI and LiDAR data, an MPT-MKD-based classification method is proposed. First, multimodal PVs are introduced, enabling the extracted features to simultaneously capture both spectral and elevation information. Next, a KD learning framework is constructed using FMs with different capacities. A larger FM is employed to obtain pseudo-supervisory information from unlabeled samples, and the class-aware predicted logits are refined to perform logits modification. Subsequently, the knowledge of the teacher model can be transferred to the student model by aligning the probability distributions of both models. This enables the effective use of a large number of unlabeled data, thereby helping to learn a more accurate class prediction model. Experimental results on the Houston 2013, MUUFL, and Augsburg datasets demonstrate the effectiveness of the proposed method. To further account for the inherent geometric and topological structures of multimodal data, incorporating graph-based methods into the construction of PVs is a highly valuable direction for future research.

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